

Soheun Yi

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Education

Carnegie Mellon University, *Ph.D., Statistics*.

Aug 2023–May 2028 (*Expected*)

Carnegie Mellon University, *M.S., Statistics*.

Aug 2023–May 2025

Seoul National University, *B.S., Mathematical Sciences*.

Mar 2017–Aug 2023

◦ *Summa Cum Laude*

◦ On leave from Fall 2019 to Spring 2021 for mandatory military service in the Republic of Korea.

Research Interests

- Statistical theory and methodology of **dynamical systems** and **optimal transport**.
- Statistics and machine learning for **scientific discovery**, particularly in computational biology.

Publications

(* for equal contribution.)

- [1] **Soheun Yi**, Y. Lu, C. Squires, and P. Ravikumar. “Concept Modulation Models: A Unified Framework for Identifiability and Extrapolation”. *arXiv preprint* (2026). arXiv: 2606.18509.
- [2] J. Lee*, **Soheun Yi***, and E. K. Ryu. “Convergence Analyses of Davis–Yin Splitting via Scaled Relative Graphs”. *SIAM Journal on Optimization* (2025).
- [3] **Soheun Yi** and E. K. Ryu. “Convergence analyses of Davis–Yin splitting via scaled relative graphs II: convex optimization problems”. *Optimization* (2025).
- [4] J. Cho, K. Sreenivasan, K. Lee, K. Mun, **Soheun Yi**, J. Lee, A. Lee, J. Sohn, D. Papailiopoulos, and K. Lee. “Mini-Batch Optimization of Contrastive Loss”. *Transactions on Machine Learning Research* (2024).
- [5] **Soheun Yi**, J. Alison, and M. Kuusela. “Toward Model-Agnostic Detection of New Physics Using Data-Driven Signal Regions”. *ML4PS Workshop at NeurIPS*. 2024. arXiv: 2409.06960.
- [6] **Soheun Yi** and S. Lee. “Filter, Rank, and Prune: Learning Linear Cyclic Gaussian Graphical Models”. *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics (AISTATS)*. 2024.

Invited and Contributed Talks

“Minimax Velocity Field Estimation”, *Contributed paper*, JSM 2026.

Aug 2026

Research Experience

Graduate Research Assistant, *Dept. of Statistics & DS*, CMU.

2023–Present

Visiting Researcher, *Deep Learning Division*, Krafton.

Apr–Jul 2023

Research Intern, *Causality Lab*, Seoul National University.

2022–2023

Research Intern, *Ryu Optimization Group*, Seoul National University.

2022

Industry Experience

Quantitative Analyst, Hyperithm, Seoul. 2019–2021

Teaching Experience

Teaching Assistant, Topics in Statistics: Deep Learning Theory. Spring 2026
Teaching Assistant, Advanced Statistical Theory. Spring 2025
Teaching Assistant, Intermediate Statistics. Fall 2024, 2025
Teaching Assistant, Advanced Data Analysis. Spring 2024
Teaching Assistant, Modern Regression. Fall 2023
Teaching Assistant, Mathematical and Numerical Optimization. Fall 2022
Deputy Leader, Team Korea at Romanian Master of Mathematics. 2019
Teaching Assistant, Korea Mathematics Olympiad Winter School. 2018

Awards and Honors

Overseas Ph.D. Scholarship, Korea Foundation for Advanced Studies. 2023–Present
Pair Merit Prize, Simon Marais Mathematics Competition. 2020
○ Top 4/150 = 3% of participants.
Undergraduate Scholarship, Korea Foundation for Advanced Studies. 2019–2023
Gold Prize, Korea Undergraduates Mathematics Competition. 2018
Presidential Science Scholarship, Korea Student Aid Foundation. 2017–2023
Finalist for International Mathematical Olympiad, Korean Mathematical Society. 2016
○ Top 13 participants in Korea.
Silver Medal, Romanian Master of Mathematics. 2016